

Vehicle rental system

In this project, I will create a relational database system for a vehicle rental company. In this system, there are 3 main types of entities: Vehicles, Customers and Staff members. The required attributes for each entity are listed below:

- Vehicles
 - License Plate (unique for each vehicle)
 - Make
 - Model
 - Model Year
 - Type of vehicle (Sedan, Minivan, SUV, Truck, etc)
 - Is the vehicle available?
 - Rented Date
 - Return Date
 - CreatedBy (employeeID)
- Customer
 - Driver's License Number (unique for each customer)
 - Name
 - Date of Birth
 - Credit/Debit Card Number (encrypt, do NOT store as plaintext)
 - Phone number (multi-valued attribute)
 - Email address (multi-valued attribute)
 - CreatedBy (employeeID)
- Staff member
 - Employee ID (unique for each customer)
 - Name
 - Date of Birth
 - Is this staff member a manager?
 - Password (encrypt, do NOT store as plaintext)
 - Company Phone number (single attribute)
 - Company Email Address (single attribute)
 - CreatedBy (employeeID)

Customers are able to rent multiple vehicles at the same time, and the system should keep track of which customers have which vehicle, as well as which employee rented the vehicle to the customer. Each vehicle is only able to be rented by one customer at a time. Using a CLI, staff members should be able to login to the system by entering their username and

password. All staff members should be able to add customers to the database by inputting the required attributes, but only managers should be able to add vehicles and other staff members. Additionally, staff members can rent vehicles to customers by inputting the customers driver's license, vehicle's license plate, and the return date (the rented date should automatically be populated with the current date). Staff members should be able to see the contents of all tables in the database (credit card numbers should be listed unencrypted, and passwords should not be returned at all), and they should also be able to see a list of only available vehicles. Additionally, staff members should be able to see a list of rented vehicles along with the name and driver's license number of the customer who rented them.